

Task 4: Setup and Use a Firewall on Windows/Linux

Use a Firewall on Windows/Linux

1. Open Firewall Configuration Tool

- **Windows**:
 - Open **Windows Defender Firewall with Advanced Security**:
 - Press `Win + R`, type `wf.msc`, and press Enter.
 - Alternatively, go to Control Panel → System and Security → Windows Defender Firewall → Advanced Settings.
- **Linux (UFW)**:
 - Open a terminal (e.g., Ctrl + T or use a terminal emulator).
 - Ensure UFW is installed: `sudo apt install ufw` (Debian/Ubuntu-based systems).
 - Verify UFW status: `sudo ufw status`.

Note: Administrative/root privileges are required for both.

2. List Current Firewall Rules

- **Windows**:
 - In **Windows Defender Firewall with Advanced Security**, select **Inbound Rules** or **Outbound Rules** from the left pane.
 - View the list of rules, including details like port, protocol, and action (allow/block).
 - For CLI, use PowerShell: `Get-NetFirewallRule | Format-Table Name,DisplayName,Enabled,Direction,Action`.
- **Linux (UFW)**:
 - Run `sudo ufw status verbose` to list active rules, showing allowed/blocked ports, protocols, and IP addresses.
 - Example output:

...

Status: active

To	Action	From
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22/tcp	ALLOW	Anywhere
80/tcp	DENY	Anywhere
...		

3. Add a Rule to Block Inbound Traffic on a Specific Port (e.g., Port 23 for Telnet)

- **Windows**:

- In **Windows Defender Firewall with Advanced Security**:

1. Click **Inbound Rules** → **New Rule**.
2. Select **Port** → Next.
3. Choose **TCP**, enter '23' in **Specific local ports** → Next.
4. Select **Block the connection** → Next.
5. Apply to all profiles (Domain, Private, Public) → Next.
6. Name the rule (e.g., "Block Telnet Port 23") → Finish.

- CLI (PowerShell, run as Administrator):

```
``powershell
```

```
New-NetFirewallRule -Name "Block_Telnet_23" -DisplayName "Block Telnet Port 23" -  
Direction Inbound -Protocol TCP -LocalPort 23 -Action Block
```

```
``
```

- **Linux (UFW)**:

- Run: `sudo ufw deny 23/tcp`

- This blocks inbound TCP traffic on port 23 (Telnet).

4. Test the Rule by Attempting to Connect to the Port

- **Locally**:

- Use a tool like **netcat** (`nc`) or **telnet**.

- On Linux/macOS: `telnet localhost 23` or `nc -zv localhost 23`.
- On Windows: `telnet 127.0.0.1 23` (if Telnet client is enabled).
- Expected result: Connection refused or timeout, confirming the port is blocked.
- **Remotely** (from another device on the same network, with permission):
 - Identify the target machine's IP (e.g., `192.168.1.100`).
 - Run: `telnet 192.168.1.100 23` or `nc -zv 192.168.1.100 23`.
 - Expected result: Connection refused or timeout.
- **Alternative**: Use `nmap` to scan: `nmap -p 23 <target\_ip>`. A "closed" or "filtered" state confirms the block.

Note: Ensure no Telnet service is running on port 23, as it's insecure and typically disabled by default.

5. Add Rule to Allow SSH (Port 22) if on Linux

- **Linux (UFW)**:
 - Run: `sudo ufw allow 22/tcp`
 - This allows inbound TCP traffic on port 22 (SSH).
 - Verify: `sudo ufw status` (should show `22/tcp ALLOW Anywhere`).
- **Windows** (if applicable, e.g., running an SSH server like OpenSSH):
 - In **Windows Defender Firewall with Advanced Security**:
 1. Click **Inbound Rules** → **New Rule**.
 2. Select **Port** → Next.
 3. Choose **TCP**, enter `22` in **Specific local ports** → Next.
 4. Select **Allow the connection** → Next.
 5. Apply to all profiles → Next.
 6. Name the rule (e.g., "Allow SSH Port 22") → Finish.
- CLI (PowerShell):

```
``powershell
```

```
New-NetFirewallRule -Name "Allow_SSH_22" -DisplayName "Allow SSH Port 22" -
Direction Inbound -Protocol TCP -LocalPort 22 -Action Allow
```

6. Remove the Test Block Rule to Restore Original State

- **Windows**:
 - In **Windows Defender Firewall with Advanced Security**:
 1. Select **Inbound Rules**.
 2. Find “Block Telnet Port 23”, right-click → **Delete**.
 - CLI (PowerShell):

```
``powershell
Remove-NetFirewallRule -Name "Block_Telnet_23"
``
```
- **Linux (UFW)**:
 - Run: ``sudo ufw delete deny 23/tcp``
 - Verify: ``sudo ufw status`` (port 23 rule should be gone).

7. Document Commands or GUI Steps Used

Below is a consolidated list of commands and GUI steps used:

- **Windows**:
 - Open Firewall: ``wf.msc`` or Control Panel → Windows Defender Firewall → Advanced Settings.
 - List Rules (CLI): ``Get-NetFirewallRule | Format-Table Name,DisplayName,Enabled,Direction,Action``.
 - Block Port 23 (GUI): New Rule → Port → TCP → 23 → Block → All profiles → Name: “Block Telnet Port 23”.
 - Block Port 23 (CLI): ``New-NetFirewallRule -Name "Block_Telnet_23" -DisplayName "Block Telnet Port 23" -Direction Inbound -Protocol TCP -LocalPort 23 -Action Block``.
 - Allow Port 22 (GUI): New Rule → Port → TCP → 22 → Allow → All profiles → Name: “Allow SSH Port 22”.
 - Allow Port 22 (CLI): ``New-NetFirewallRule -Name "Allow_SSH_22" -DisplayName "Allow SSH Port 22" -Direction Inbound -Protocol TCP -LocalPort 22 -Action Allow``.
 - Remove Rule (GUI): Inbound Rules → Find “Block Telnet Port 23” → Delete.
 - Remove Rule (CLI): ``Remove-NetFirewallRule -Name "Block_Telnet_23"``.
- **Linux (UFW)**:

- Open UFW: Terminal, check status with ``sudo ufw status``.
- List Rules: ``sudo ufw status verbose``.
- Block Port 23: ``sudo ufw deny 23/tcp``.
- Allow Port 22: ``sudo ufw allow 22/tcp``.
- Remove Rule: ``sudo ufw delete deny 23/tcp``.

8. Summarize How Firewall Filters Traffic

A firewall filters network traffic by enforcing rules that control which packets are allowed or blocked based on criteria like:

- **Source/Destination IP**: Specifies which devices can send/receive traffic.
- **Port Number**: Determines which services (e.g., port 23 for Telnet, 22 for SSH) are accessible.
- **Protocol**: Filters by protocol type (e.g., TCP, UDP).
- **Direction**: Manages inbound (incoming) or outbound (outgoing) traffic.
- **Action**: Allows, blocks, or redirects traffic.

How It Works:

- The firewall inspects packet headers against its rule set.
- Rules are processed in order (or priority). The first matching rule determines the action (allow/block).
- If no rule matches, the default policy (e.g., deny all) applies.
- Example: Blocking port 23 prevents Telnet connections, while allowing port 22 enables SSH access.

Outcome: These tasks demonstrate basic firewall management skills, including rule creation, testing, and documentation, and provide an understanding of how firewalls secure networks by filtering traffic.
